

TRAVERSE CITY CHERRY CAPITAL, MI, USA

WMO: 726387

Lat: 44.741N

Lon: 85.583W

Elev: 188

StdP: 99.08

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14850

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.7	-15.0	-21.3	0.6	-16.9	-18.6	0.7	-14.2	10.6	-6.0	9.3	-5.7	2.7	280	0.537

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.3	31.5	21.9	29.9	21.2	28.2	20.3	23.6	29.0	22.6	27.6	21.7	26.4	4.3	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.0	17.1	26.8	21.0	16.0	25.2	20.0	15.0	24.6	71.4	29.6	67.3	27.4	63.9	26.3	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.1	8.2	7.4	DB	-21.4	34.5	4.1	1.5	-24.4	35.5	-26.8	36.4	-29.1	37.3	-32.0	38.4	(4)
(5)			WB	-21.7	25.4	4.0	1.2	-24.6	26.3	-26.9	27.0	-29.2	27.7	-32.1	28.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.2	-5.2	-4.7	-0.1	6.2	12.7	18.5	21.2	20.4	16.7	10.1	3.6	-1.7
	DBStd	10.54	5.43	5.87	6.21	5.07	4.96	4.24	3.43	3.14	4.31	4.82	4.65	4.79
	HDD10.0	1989	471	412	319	137	25	1	0	0	2	59	199	364
	HDD18.3	3979	729	646	571	365	187	48	12	14	82	261	442	622
	CDD10.0	1338	0	1	7	23	111	257	346	323	203	61	7	1
	CDD18.3	285	0	0	1	1	14	54	100	79	33	4	0	0
	CDH23.3	2537	0	0	8	19	174	532	910	611	250	34	0	0
	CDH26.7	739	0	0	2	3	51	176	284	158	61	6	0	0
Wind	WSAvg	3.1	3.5	3.5	3.3	3.4	2.9	2.6	2.7	2.5	2.6	3.1	3.4	3.3
Precipitation	PrecAvg	771	60	36	46	60	62	79	72	74	100	68	64	55
	PrecMax	995	271	105	185	137	146	234	248	198	233	181	119	107
	PrecMin	565	25	7	4	22	8	9	7	13	9	23	10	22
	PrecStd	111	44	22	34	28	31	51	45	42	56	38	26	24
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.8	12.3	22.2	26.1	31.0	33.3	33.9	32.6	31.3	27.2	20.0	13.4
		MCWB	7.9	7.9	14.6	15.9	20.5	23.3	23.6	22.5	21.6	19.5	13.4	11.5
	2%	DB	6.3	8.3	16.0	22.2	27.8	30.9	31.4	30.2	28.2	23.6	15.7	8.9
		MCWB	4.3	5.3	10.8	13.7	18.6	21.0	22.3	21.8	20.2	17.6	11.5	6.7
	5%	DB	3.8	5.1	12.0	18.2	24.9	28.7	29.7	28.2	26.2	20.9	13.0	6.6
		MCWB	2.4	2.9	7.9	10.8	16.7	20.2	21.5	20.8	19.4	15.7	9.4	4.5
	10%	DB	2.3	3.0	8.4	15.0	21.9	26.5	28.0	26.8	24.1	17.8	10.9	4.4
		MCWB	1.2	1.2	4.9	9.3	15.2	19.1	20.5	19.8	18.3	14.0	7.9	2.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.4	9.1	15.8	17.6	22.1	24.4	25.6	24.5	23.0	21.2	14.7	11.7
		MCDB	9.3	10.7	21.6	23.3	28.9	31.4	30.8	29.6	28.2	25.5	18.1	12.8
	2%	WB	4.3	5.3	11.9	14.6	19.8	22.5	24.0	23.1	21.7	18.6	12.2	7.4
		MCDB	5.8	7.9	15.1	20.1	25.5	28.9	29.3	27.8	25.8	22.2	15.0	8.6
	5%	WB	2.5	3.1	7.9	12.0	17.9	21.3	22.7	22.1	20.6	16.2	10.1	4.7
		MCDB	3.7	4.8	11.0	17.3	23.3	26.6	27.7	26.6	24.7	19.8	12.9	6.3
	10%	WB	1.2	1.4	5.2	9.5	16.1	20.1	21.7	21.1	19.4	14.2	7.9	2.9
		MCDB	2.3	3.0	8.3	14.5	21.0	24.9	26.3	25.0	23.4	17.6	10.4	4.3
Mean Daily Temperature Range	5% DB	MDBR	6.3	7.8	9.2	11.1	12.1	11.6	11.3	10.6	10.6	9.0	6.9	5.6
		MCDBR	7.7	9.6	13.4	17.1	15.6	13.9	13.0	12.4	12.7	12.9	10.8	8.4
		MCWBR	6.6	7.4	9.1	10.0	8.2	6.6	6.0	5.8	6.3	7.5	7.6	7.1
	5% WB	MCDBR	7.1	8.9	12.1	15.3	13.8	11.7	10.7	10.1	10.6	11.0	9.8	7.8
		MCWBR	6.4	7.1	9.0	10.0	8.2	6.4	5.6	5.5	6.1	7.8	7.7	7.1
Clear-Sky Solar Irradiance	taub	0.284	0.295	0.321	0.355	0.384	0.402	0.397	0.387	0.372	0.341	0.308	0.283	
	taud	2.380	2.327	2.391	2.349	2.332	2.308	2.341	2.364	2.396	2.479	2.506	2.438	
	Ebn at Noon	837	892	915	906	885	868	867	864	848	832	807	803	
	Edh at Noon	75	95	103	117	123	127	121	114	102	82	67	65	
All-Sky Solar Radiation	RadAvg	1.20	2.12	3.55	4.55	5.54	6.05	6.10	5.19	4.05	2.39	1.32	0.89	
	RadStd	0.11	0.22	0.40	0.44	0.48	0.42	0.28	0.34	0.37	0.29	0.19	0.10	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page